The six-card cover problem: an overview

The case $r = 6$ of the Erdős–Lovász cover number problem.

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1. The question

Erdős and Lovász asked in 1974, in the paper that introduced the local lemma, for the minimum number of edges $g(r)$ in an r-uniform intersecting hypergraph whose cover number is r. Erdős later called it one of his three favourite combinatorial problems and offered $500 for a linear upper bound; Kahn proved one in 1994 (*J. Amer. Math. Soc.* 7, 125–143). The exact values known are $g(3) = 6$, $g(4) = 9$ and $g(5) = 13$, the first two due to Tripathi (arXiv:1409.4610) and the last to Barát and Wanless (arXiv:2011.04444; *J. Combin. Des.* 29 (2021) 260–286), and the strongest published general lower bound, $g(r) \geq ((41 - \sqrt{19})/12 - o(1))r$ (arXiv:2606.24878), gives only $g(6) \geq 14$. This overview concerns $r = 6$. What the literature does and does not already settle is §6, and it should be read before the value here is quoted anywhere. [P5] and [P6] turn from $r = 6$ to every r. For $r = 7$, [P5, §1] records an explicit 21-card family with cover number 7, so $18 \leq g(7) \leq 21$.

A family of finite sets is **6-uniform** if every member has exactly six elements, and **pairwise intersecting** if any two members share an element. We call a member a *card* and an element a *symbol*.

A **transversal** (or cover) is a set of symbols meeting every card. The smallest size of a transversal is the transversal number, written τ.

In a pairwise intersecting 6-uniform family, the six symbols of any single card already meet every other card. So $\tau \leq 6$ always, and $\tau = 6$ is the same statement as *no five symbols cover every card*.

The quantity studied here is

$$g(6) = \min\{|F| : F \text{ is 6-uniform, pairwise intersecting, } \tau(F) = 6\}.$$

2. The chain

stage	statement	where
at most 15 cards	impossible	[P1] and [P2], two independent routes
exactly 16 cards	impossible	[P3], computer-assisted
exactly 17 cards	an explicit family exists	[P4]

The first two give $g(6) \geq 17$. The third gives $g(6) \leq 17$. Together,

$$g(6) = 17.$$

3. The papers

[P1] Fifteen cards by hand: the histogram route. The original argument, and a hand proof of $g(6) \geq 16$. Symbol degrees are forced into $\{2, 3, 4\}$; a finite lemma about edge covers of K_7 is proved by hand and also checked exhaustively; a weighted counting inequality cuts the possible degree histograms down to thirteen, by a short count; each of the thirteen is then eliminated by hand.

[P2] The eleven-card lemma. A later and shorter route to the same place. It proves that *every* pairwise intersecting 6-uniform family of eleven cards has a transversal of size four, and then gets the fifteen-card exclusion in two lines. The eleven-card lemma is also what forces the maximum degree to be exactly four in the sixteen-card stage, so [P2] is used by [P3] while [P1] is not.

[P3] Sixteen cards. Deleting the four cards of a degree-four symbol leaves a twelve-card kernel with $\tau = 5$. That kernel either has a second degree-four symbol or does not, and the two cases are handled separately. If it does not, the kernel is one of 341 objects of maximum degree three, enumerated and excluded one by one. If it does, deleting that symbol's cards leaves an eight-card core. The cores are enumerated up to relabelling — there are 463 — and for each one the completion problem is written as a CNF formula. All 463 are unsatisfiable, and each unsatisfiability has a DRAT certificate accepted by an independent checker.

[P4] A seventeen-card witness. Seventeen cards on 27 symbols, with all $\binom{17}{2} = 136$ pairs intersecting and none of the $\binom{27}{5} = 80{,}730$ five-symbol sets a transversal. This is a direct finite check and needs nothing from the other three papers.

[P5] Every r. Proposed proofs, in the framework of Sivashankar (arXiv:2606.24878): the residual cover bound with constant $+3$, which gives $g(r) \geq 3r - 3$ for every r, and the exclusion of equality, which gives $g(r) \geq 3r - 2$ for $r \geq 16$. Its finite steps are checked exhaustively; its other steps are unreviewed.

[P6] Large r. A proposed proof of a new asymptotic lower bound, $\liminf g(r)/r \geq 3.1108$, above the constant 3.0534 of that preprint, with the final minimisation certified by exact rational arithmetic. At $r = 6$ these papers give only $g(6) \geq 15$, so they add nothing to the value above.

4. What carries the weight

The two bounds are not of the same kind, and it is worth being plain about that.

The upper bound is a witness. Anyone can check it with a short script and no libraries: seventeen sets, 136 pairs, 80,730 subsets. Nothing is trusted except arithmetic.

The fifteen-card stage has a hand proof in [P1]; its K_7 lemma is also checked exhaustively (294,239,817 cases). Beyond that the lower bound is computer-assisted at two points. The eleven-card lemma's third branch reduces by hand to a short statement about 3-element sets on eleven points, checked by a SAT solver in seconds; the earlier exhaustive search over 5,373 instances is kept as a second check. The sixteen-card exclusion rests on the reduction to 463 cores and on 463 unsatisfiability results.

For the last of those, the solver's word is not the evidence. Each CNF has a DRAT proof, and a DRAT proof checked by `drat-trim` removes the SAT solver from the trusted base. What a DRAT proof does **not** establish is that the reduction reaches every possible counterexample, or that the CNF says what the mathematics says. Those two obligations live in the reduction source and the encoding source, and they are the parts a reader should read rather than run. `AUDIT.md` records what an independent re-run did and did not settle. Each computation of the sixteen-card stage has also been repeated by separately written code — the cores from their definition, the encoding without symmetry breaking, the extension test by SAT ([P3, §7]); that repeats the computations, not the arguments.

5. Citation convention

The papers of this set are cited as [P1] to [P6]. A bare bracketed number inside a paper is an entry in that paper's own reference list. Each paper numbers its own results from 1.

tag	file
[P1]	`paper_1_fifteen_cards_histogram_route.md`
[P2]	`paper_2_eleven_card_lemma.md`
[P3]	`paper_3_sixteen_cards.md`
[P4]	`paper_4_seventeen_card_witness.md`
[P5]	`paper_5_every_r.md`
[P6]	`paper_6_asymptotic.md`

6. Literature boundary

Two sources were reviewed: a general lower bound and an eighteen-card construction (see the reference lists of [P1] and [P3]). Neither settles whether the value obtained here is already known. A construction that is minimal inside a particular projective plane is not the same thing as the unrestricted value of $g(6)$, and should not be quoted as such. No priority claim is made from the computation alone; a proper literature and priority review should come before any journal submission.

A wider search was made during the second verification pass and is recorded in `AUDIT.md` §4.7. It found the exact values $g(3) = 6$, $g(4) = 9$ and $g(5) = 13$, and no source determining $g(6)$; it also confirmed the distinction drawn above against the primary text, since the eighteen-edge example of Barát and Wanless is the minimum inside $PG(2, 5)$ and is a value of the projective-plane restricted function, not of $g(6)$. That search was a search. It cannot establish that nothing was missed, and the obligation stated in the previous paragraph is unchanged.

7. Status, priority, and the use of AI assistance

This is a research draft, not a settled result. No priority is claimed for the value obtained here, or for any lemma in this set. The literature was reviewed only as far as the two sources named in the reference lists, which do not settle whether the value is already known.

The lower bound has not been independently reviewed. It is computer-assisted at three points, and while DRAT certification removes the SAT solver from the trusted base, two obligations remain open by construction: that the finite reduction reaches every possible counterexample, and that the CNF encoding means what the mathematics means. Neither is discharged by any certificate. No human referee and no proof assistant has checked this work. Until one has, the right reading of the lower bound is *a reproducible computation whose reductions are written down and open to review*, not *a theorem*. The upper bound is different in kind: it is a witness, and anyone can check it in a few seconds.

AI assistance was used throughout, including in the review. Large language models were used as tools: ChatGPT (OpenAI) for algebraic derivation, for much of the code in `code/`, and for drafting; Claude (Anthropic) for independent re-running and auditing of the computations, for the encoding controls, for the reconstruction of the lost enumeration step, and for review and prose. Where the two disagreed, computation settled it and the text records the resolution. The research direction, the objects studied, and the responsibility for every claim are the author's.

The same tools were used to check the work, so the checking carries the same caveat as the work. What the reader is asked to rely on is not any assurance about the tools but the discipline behind them: **every number reported here is regenerated by a script in `code/` or certified by a proof file in `results/`, and every script has been read.** No statement in these papers rests on the assertion of a model. `AUDIT.md` records what an independent re-run settled and what it could not.